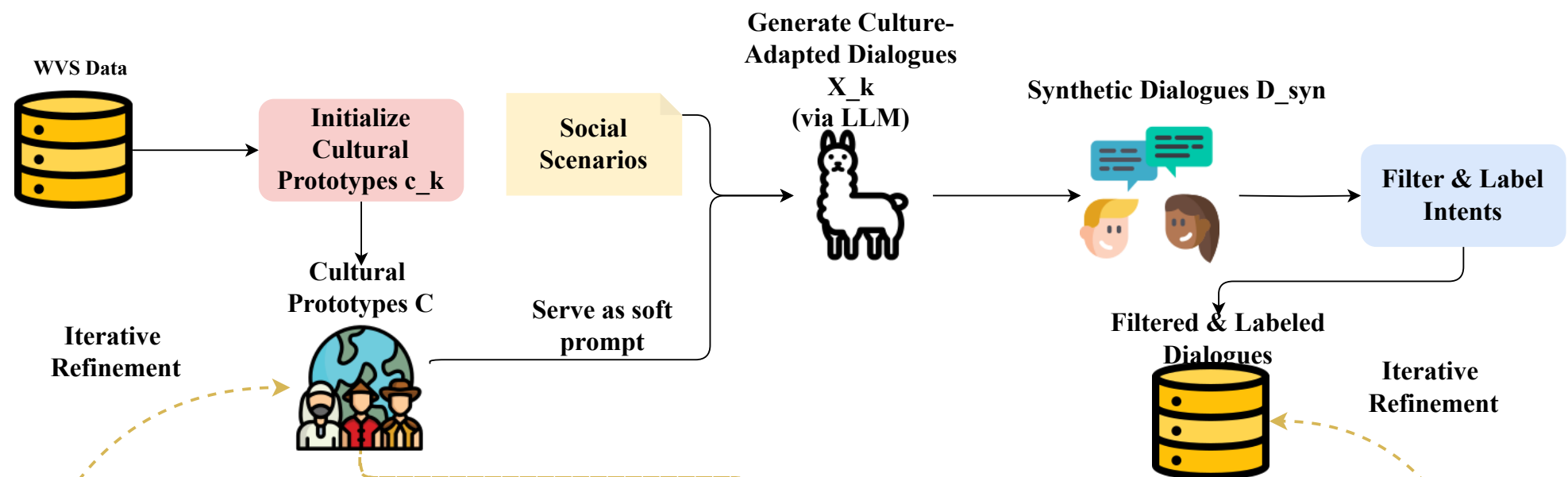
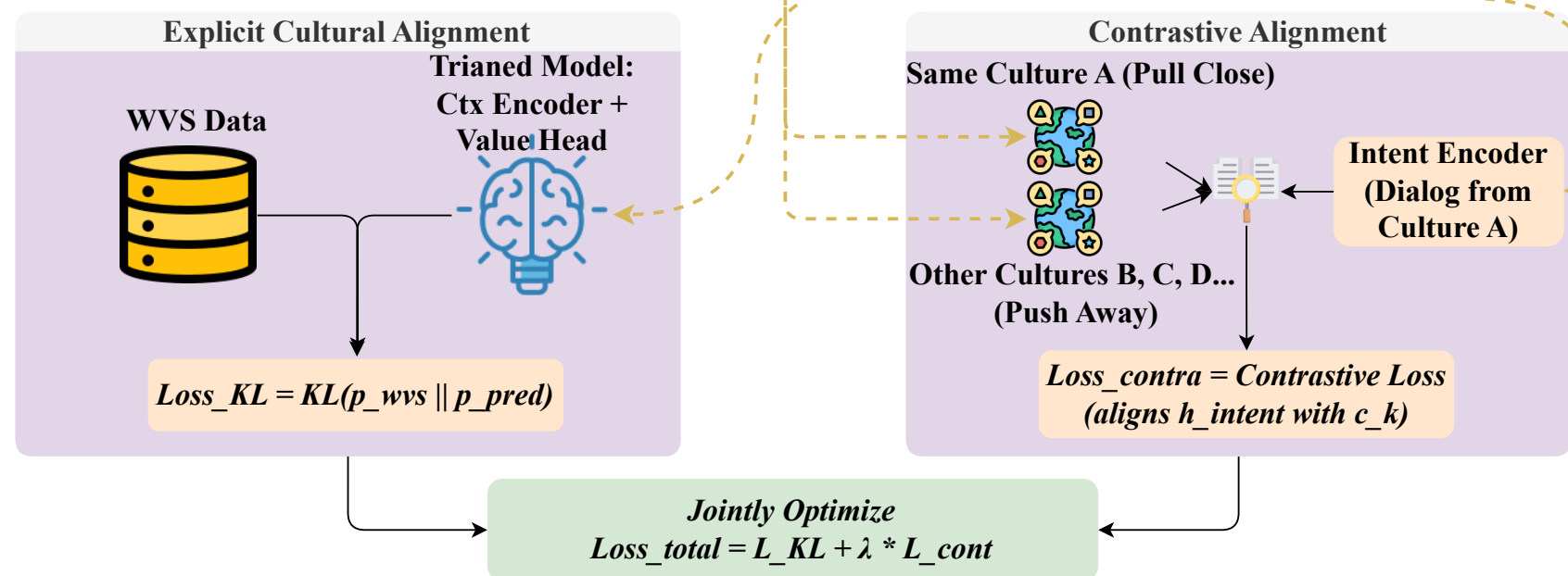


Phase 1: Data Synthesis & Preparation



Phase 2: Unified Contrastive Training (ProCA Core)



Phase 3: Iterative Refinement & Evaluation

